

Then, of course, there's Ethernet. In 1973, Robert Metcalfe's PhD thesis at Harvard outlined the idea. The same year, after he joined Xerox's PARC as a researcher, he was required to build a networking system for the Center's computers. The company wanted the network because they were building the first laser printer, and wanted to be able to share the printer amongst all PARC computers. Necessity is the mother of... Metcalfe, therefore, had two things to achieve: a fast enough network to drive the fast printer, and a connectivity system that could scale up to hundreds of computers.

It is "generally considered" that Ethernet was invented in May of 1973, but Metcalfe claims Ethernet was invented gradually over several years. We don't need to tell you how much this landmark invention meant: in three words, it revolutionised networking.

Coming to processors, Intel's 8008, in 1972, was the first 8-bit microprocessor. It was soon replaced by the 8080, in 1974, which was the beginning of a revolution. Michael Kanellos of CNET News.com sums it up: "The breakthrough moment for microprocessing came in 1974, according to many, with the 8080 processor. Not only did the chip feature a more complex instruction set, it came in a package with 40 pins, two innovations that greatly expanded its capabilities." The transition to 16-bit happened in 1978: Intel's 8086 saw the light that year.

From processors to personal computers: the story is long and complex, and is chock-full of "generally believed"—you'd do well to visit [www.blinkenlights.com/pc.shtml](http://www.blinkenlights.com/pc.shtml) for a lowdown on which "The First Personal Computer" was. But here, we should mention the French Micral, arguably the first

commercial, non-kit, personal computer based on a microprocessor, the Intel 8008. Now that's a lot of qualifications, but you'll notice that it is a first of sorts. Then in 1975 came the Altair 8800, as a kit. Some call it the first personal computer; some call it the first microcomputer. Some say it's neither. Whatever the case, it was a first of some sort! Specs: 2 MHz Intel 8080, 256 bytes RAM.

Enter Apple. The two Steves—Jobs and Wozniak—designed and built the Apple I in 1976. It wasn't easy. Jobs first had to persuade a computer store to buy his creation, and the store ended up ordering 50 units at \$500 each. He then went about obtaining the components. This must not have been fun: he had to sell some of his stuff to buy them. And after the parts were procured, Wozniak and a friend called Ronald Wayne assembled the Apple I. It was the first personal computer with a keyboard and which ran on a microprocessor, while also having a connection to a monitor. Too many qualifications, and a better reason for mentioning it is probably that the Apple I was the first personal computer sold in fully-assembled form.

Succeeding the Apple I was—no prizes—the Apple II: it was a hit, partly because it was capable of being hooked up to a colour TV set. Colour and high-res graphics were a first, as were its sound capabilities. Some say (and some don't) that the Apple II was "the beginning of the personal computer revolution." Which, actually, many things were—take your pick!

Other highlights of the decade included the development of the inkjet as well as the laser printer. By HP, right? In fact, that's what even President G H W Bush believed: in a certain speech, he congratulated HP engineers for

having invented the laser printer. HP actually came up with the LaserJet in 1984. It's Xerox that invented the laser printer. (Perhaps we should qualify that with a "generally considered.")

People discovered home entertainment—and theatres lost their monopoly on movie lovers—when the VCR was invented in 1970. In 1972, Seymour Cray founded Cray Research and introduced the first successful commercial supercomputer, and megalomaniacally named it the Cray I. And in 1977, Bell Labs built a prototype cell phone system in Chicago, and tested the system the next year. The year after that, the first commercial cell phone system was started in Tokyo. Last but not least—can we leave out mention of Microsoft, founded in 1977 by Bill Gates and Paul Allen?

As we move on to the 80s, we note that much of what we're familiar with today had been developed by 1979, at least in rudimentary form. There was colour graphics, there was 16-bit processors, there was the hard disk. The stage was set for everyman's PC. ■

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## ARPANET

Formally, ARPANET was the first wide-area packet-switching network. Much of its development happened in the 70s, and we think no-one will dispute the notion that nothing more significant than this happened in the decade—processors and PCs notwithstanding: no other *single* development contributed as much to the computing experience as we know it today.

ARPANET was to serve as a test-bed for networking technologies. The plan required the network to be composed of computers known as Interface Message Processors (IMPs). The IMPs at each node performed packet-switching functions, and used modems connected to leased lines (the initial speed was 50 kbps). ARPANET initially ran on the NCP-Network Control Program, or Network Control Protocol. In 1983, TCP/IP replaced NCP.

To read up on the history of ARPANET, a particularly good site is [www2.dei.isep.ipp.pt/docs/arpa.html](http://www2.dei.isep.ipp.pt/docs/arpa.html).

